Yijie Ma

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Skills

C/C++, Go, Python, Distributed Systems, gRPC, API Gateway/L7 Proxy, AI Infrastructure (MCP, A2A, RDMA), Kubernetes, OpenTelemetry, Networking, Storage & Filesystems, Desktop App Development (Linux/macOS/Windows)

Work Experience

Google April 2017 - Present

Software Engineer

Google Cloud API Gateway & Service Infrastructure

- Architected C++ OpenTelemetry distributed tracing and W3C Trace Context propagation across HTTP and gRPC for Cloud, GenAI, Model Context Protocol (MCP), and Agent-to-Agent (A2A) APIs.
- Led delivery of the Express Stack—a fast-track API serving tier decoupling rapid AI/Cloud feature releases from the 99.999%-SLO core gateway—architecting C++ dual-identity config resolution for zero-friction onboarding.
- Pioneered multi-agent workflows for release automation using reusable AI Skills to evaluate 200+ code changes across a 7-dimension risk rubric, generate release reports, and automate staging promotions.
- Mentored junior engineers via 1:1 architecture walkthroughs, milestone scoping, and code reviews to deliver Agent-to-Agent (A2A) telemetry and end-to-end prober suites.

Google Kubernetes Engine (GKE) Networking

- Engineered GKE control-plane automation in Go to auto-provision multi-NIC RDMA network topologies (VPCs, subnets, IP ranges) for AI-optimized GPU clusters.
- Designed concurrent resource provisioning pipelines and validated multi-node GPUDirect-RDMA performance on NVIDIA H200 GPU clusters using NVIDIA's NCCL test suite.

gRPC Open Source Remote Procedure Call Framework

- Core developer and Maintainer of gRPC (40K+ GitHub stars) with 150+ authored PRs across C++ core, networking, and observability.
- Designed and built the cross-platform EventEngine asynchronous DNS resolver across Linux/POSIX and Windows.
- Co-authored gRFC A79; designed and built the high-performance Metrics framework with C++ template technique and the OpenTelemetry plugin.
- Designed the POSIX `fork()` handler system and resolved critical concurrency deadlocks, security issues, and performance bottlenecks for millions of users.

Google Drive Desktop Application

- Built rsync-style differential upload and Gzip stream compression encoders with compressibility heuristics to save upload bandwidth for millions of users.
- Built the USB device backup subsystem backing up millions of files daily and increased backup speed by 10x (140s to 18s for 1GiB) by optimizing I/O buffers.
- Built the cross-platform Preferences module with user-level and admin-level overrides, and refactored macOS/Windows UI controllers to delete 4,000+ lines of C++ code.
- Added Linux Crashpad and Windows heap-stack crash reporting (capturing 1,000+ daily crashes), mitigated a live outage for 10,000+ users, and delivered 100+ bug fixes.

Stadia Cloud Gaming Service

- Built a Linux Cgroup library for setting up container resource configurations on cloud gaming VMs.
- Increased game click-to-play launch speed by 75% (1.58s to 396ms) via CPU profiling and container setup optimization.
- Built a custom Windows shell to replace `explorer.exe` with AppLocker lockdown and multi-process game lifecycle management.

NetApp August 2015 - March 2017

Member of Technical Staff

Worked on the AltaVault Cloud-Integrated Storage device. Built a filesystem software and used RocksDB for cataloging users' snapshot data. Built full or incremental backup and restore functionalities.

Education

Carnegie Mellon University Sep 2013 - May 2015

Master of Science in Electrical and Computer Engineering

Coursework: Introduction to Computer System, Embedded Realtime System, Advanced Storage System, Internet Service, Cloud Computing, Distributed System

Donghua University Sep 2009 - May 2013

Bachelor of Engineering in Electrical Engineering and Automation

Awards: 2012 IEEE-standard Micromouse competition national award issued by China Computer Federation (CCF)